Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The DNS protocol used UDP to communicate with the DNS server and obtain the IP address for the domain yummyrecipesforme.com. In each log event, the first two lines show the UDP request sent from the browser to the DNS server. The third and fourth lines display an ICMP error response stating “udp port 53 unreachable,” which indicates that the DNS server could not be reached. Since port 53 is specifically used for DNS traffic, this suggests a problem with the DNS service. The presence of the plus sign after the query ID 35084 and the “A?” flag further confirms that the system was attempting to perform a DNS lookup operation. Based on the repeated ICMP error messages and the failed DNS queries, it is highly likely that the DNS server was unavailable or not responding.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
At 1:24 p.m. today, customers reported receiving a “destination port unreachable” message while trying to access yummyrecipesforme.com. The cybersecurity team responsible for providing IT support to the client organization has started investigating the issue to restore website access. During the investigation, packet capture analysis was performed using tcpdump, and the logs revealed that DNS port 53 was unreachable. The next step is to determine whether the DNS server is offline or if firewall rules are blocking traffic on port 53. Possible causes include a Denial of Service (DoS) attack affecting the DNS server or an incorrect server or firewall configuration.